

Yue Liu

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SUMMARY OF QUALIFICATIONS

- **Applied AI & Data Architecture:** Bridging a strong foundation in high-throughput database systems and applied artificial intelligence to build robust data-to-model pipelines.
- **Uncertainty Quantification (UQ) & Math:** 4 years of professional engineering experience heavily grounded in advanced probability. Highly proficient in utilizing stochastic processes, Bayesian hierarchical models, and Copulas to model the probabilistic boundaries of complex physical systems.
- **High-Volume ETL Operations:** Proven track record of designing automated workflows to ingest, process, and reconcile massive, multi-dimensional datasets for large-scale industrial networks.

TECHNICAL SKILLS

Mathematics & Analytics: Uncertainty Quantification (UQ), Bayesian Hierarchical Models, Copulas, Stochastic Process Time-Series, Monte Carlo Simulations.

Programming & AI: Python, Java, SQL, C, MATLAB, Semantic Segmentation (U-Net), Custom Data Labeling (Fiji/ImageJ).

Data & Systems Architecture: Relational Database Management Systems, High-throughput ETL Pipelines, Operating Systems Concurrency, TCP/IP Protocols.

EDUCATION

Bachelor of Computer Science (Second Degree)

Sept 2024 – Present

University of Calgary – Calgary, AB

- **CS Major GPA:** 3.85 / 4.0
- **Data & AI:** Database Management Systems (CPSC 471: A+), Intro to Artificial Intelligence (CPSC 383: A+).
- **Systems & Networks:** Principles of Operating Systems (CPSC 457: A+), Computer Networks (CPSC 441: A), Computing Machinery I (CPSC 355: A+).
- **Upcoming:** Machine Learning (CPSC 544).

Master of Engineering Science

2019 – 2021

Western University – London, ON

- **Grade:** 92.5 / 100
- **Thesis:** Integrity Management of Oil & Gas Pipeline.
- Developed highly reliable statistical models using Monte Carlo Simulations and advanced probability methods to predict complex infrastructure failure risks under environmental uncertainties.

Bachelor of Engineering (Civil Engineering)

2012 – 2016

Southeast University – Nanjing, China

PROFESSIONAL EXPERIENCE

JANA Corporation

Remote / Aurora, ON

Data Analyst II / Model Developer (Co-op/Contract)

May 2025 – Aug 2025, May 2026 – Present

- Design and optimize robust data processing workflows for backend teams, handling high-volume operational datasets and multi-modal asset tracking metrics.
- Collaborate with cross-functional software teams to implement scalable, data-driven quantitative analysis frameworks for complex industrial networks.

Integrity Engineer (P.Eng. designation in progress)

Oct 2021 – Dec 2024

- Led the development of automated ETL processes and quantitative models, reconciling discrepancies across massive datasets for dynamic physical systems.
- Applied rigorous statistical intuition to model degradation risks and probabilistic boundaries, utilizing script-based Python frameworks to seamlessly integrate raw sensor and inspection data into the central architecture.

RESEARCH & ACADEMIC PROJECTS

Semantic Segmentation for Microscopy Images

July 2026 – Present

Independent Collaboration

- Applied data-to-model workflow principles to a computer vision task by engineering a custom data-labeling pipeline using Fiji.
- Established strict Standard Operating Procedures to extract ground-truth datasets from raw chemical samples, and successfully trained a U-Net deep learning model in Python to perform semantic segmentation of graphene layers.

PUBLICATIONS

"Reliability analyses of corroding pipelines using different approaches to characterize uncertainties in internal pressure", Journal of Pressurized Vessel Technology, 2021.